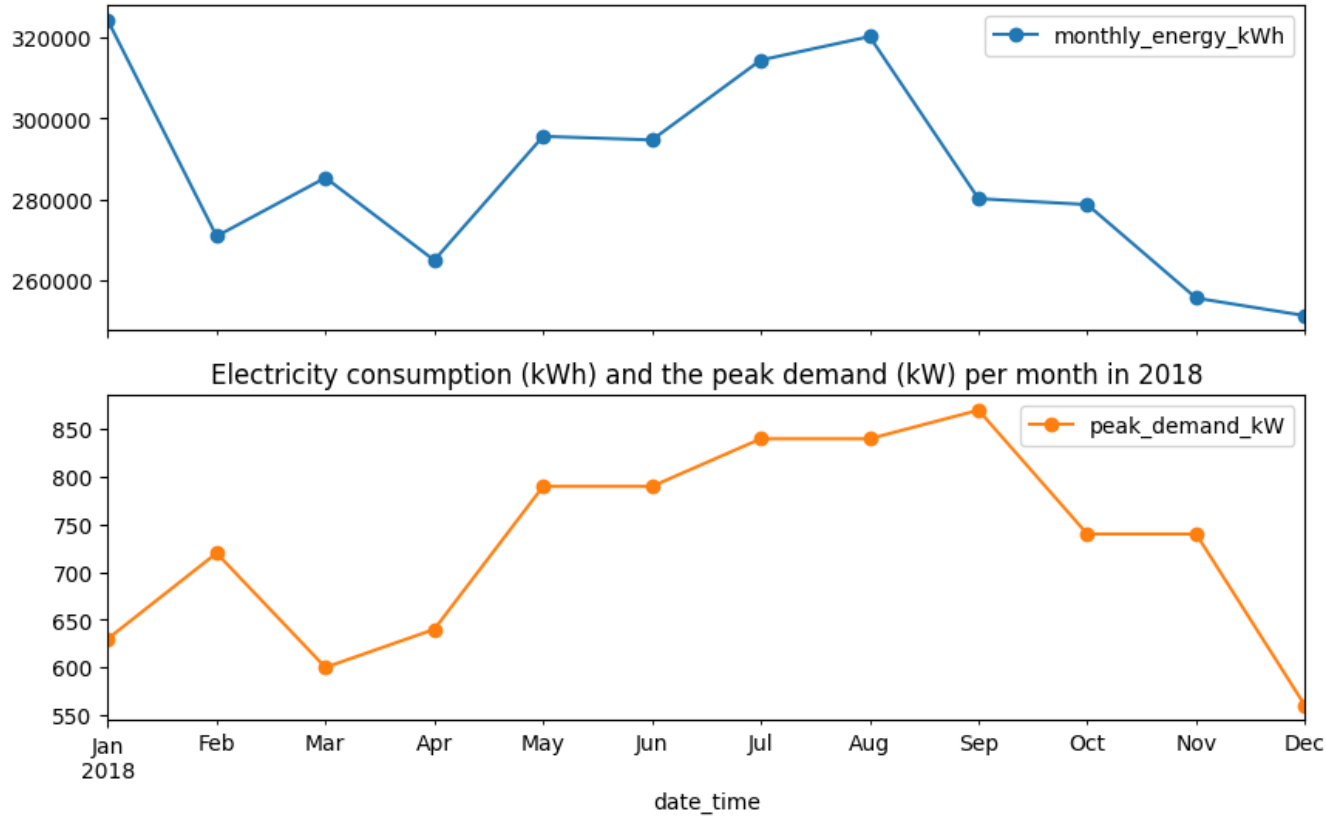
Energy Forecasting Model

Technical Report — GRU-Based Demand & Consumption Prediction

1. Understanding building electricty load

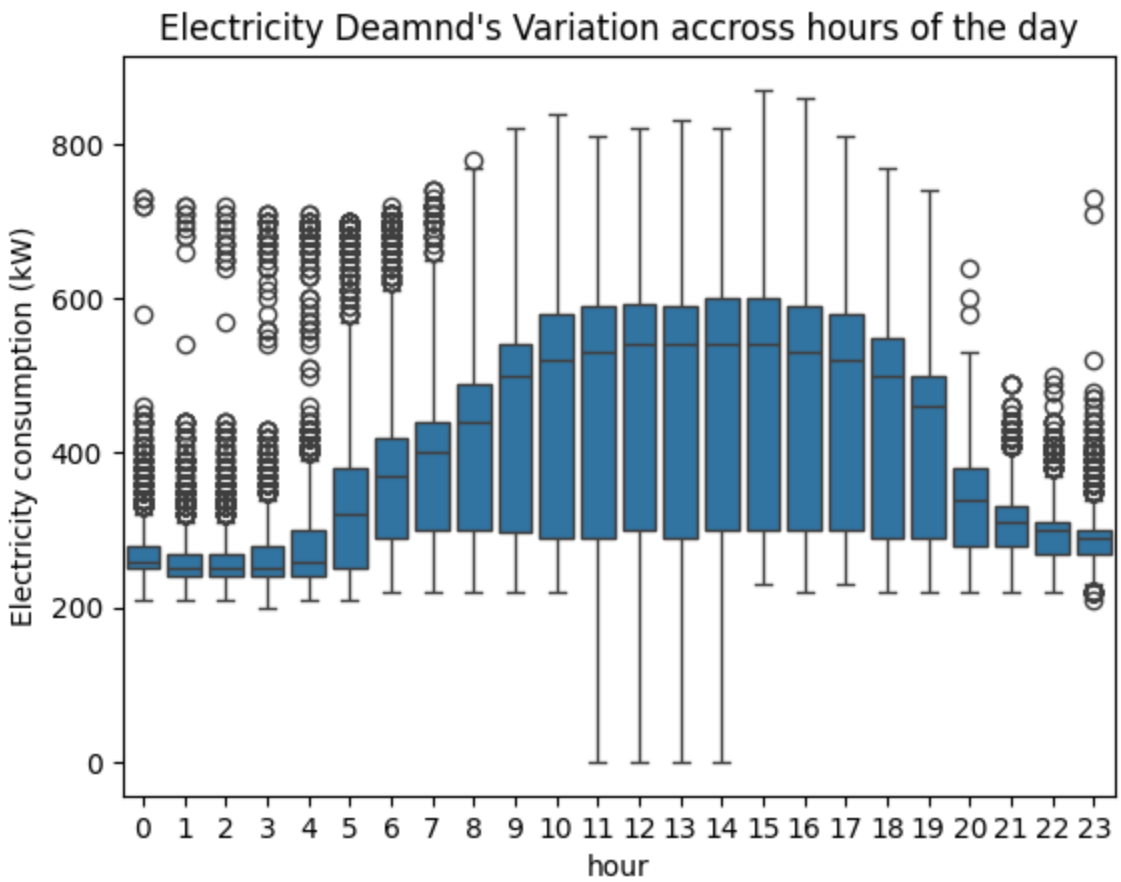
What is the electricity consumption (kWh) and the peak demand (kW) of the building per month in 2018?



Electricity consumption is highest in summer (July–August) and lowest in winter (November–December). Peak demand follows a similar pattern, reaching its maximum in September, indicating higher load stress during late summer.
This difference highlights the distinction between energy usage and instantaneous demand.

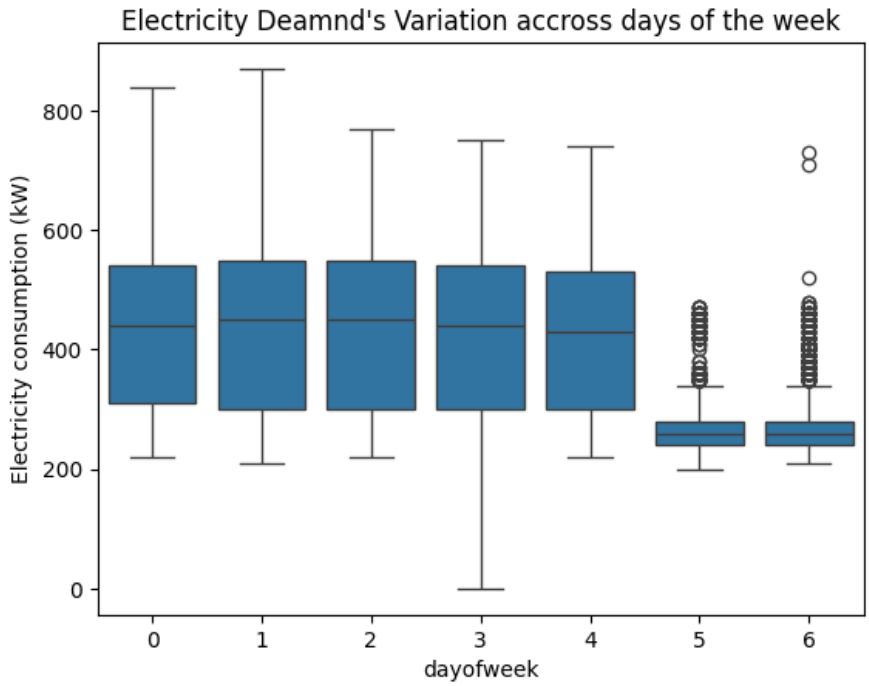
1.b) How does electricty demand vary

i) across hours of the day?



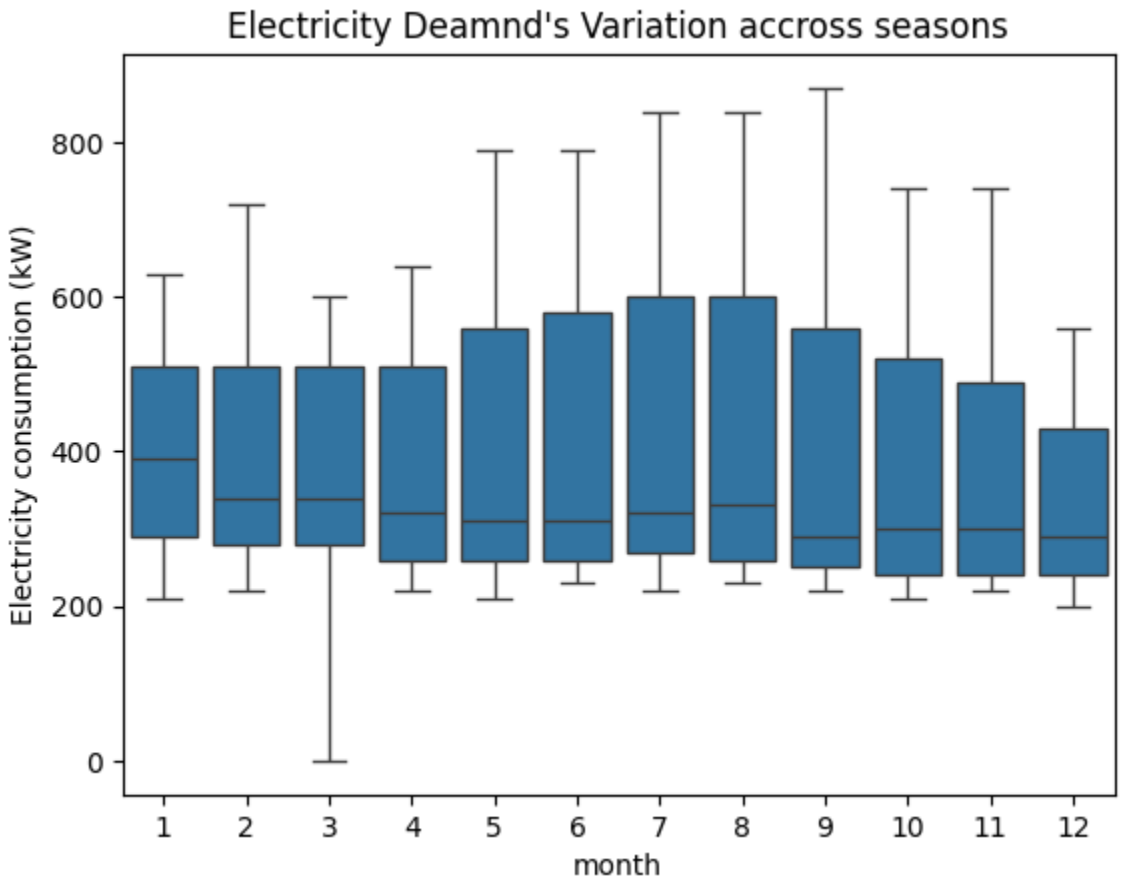
Electricity demand is lowest during late night and early morning (00:00–05:00 a.m.), rises steadily from early morning (06:00–09:00 a.m.), peaks around midday to mid-afternoon (11:00–16:00 p.m.), and then declines gradually in the evening, returning to lower levels late at night.
This analysis highlights the typical daily load profile and the presence of base load and peak demand hours.

ii) across days of the week?



Electricity demand is higher and more variable on weekdays (Monday–Friday), with similar median levels, and significantly lower on weekends (Saturday–Sunday), where demand is more stable and consistently reduced.

iii) across seasons?

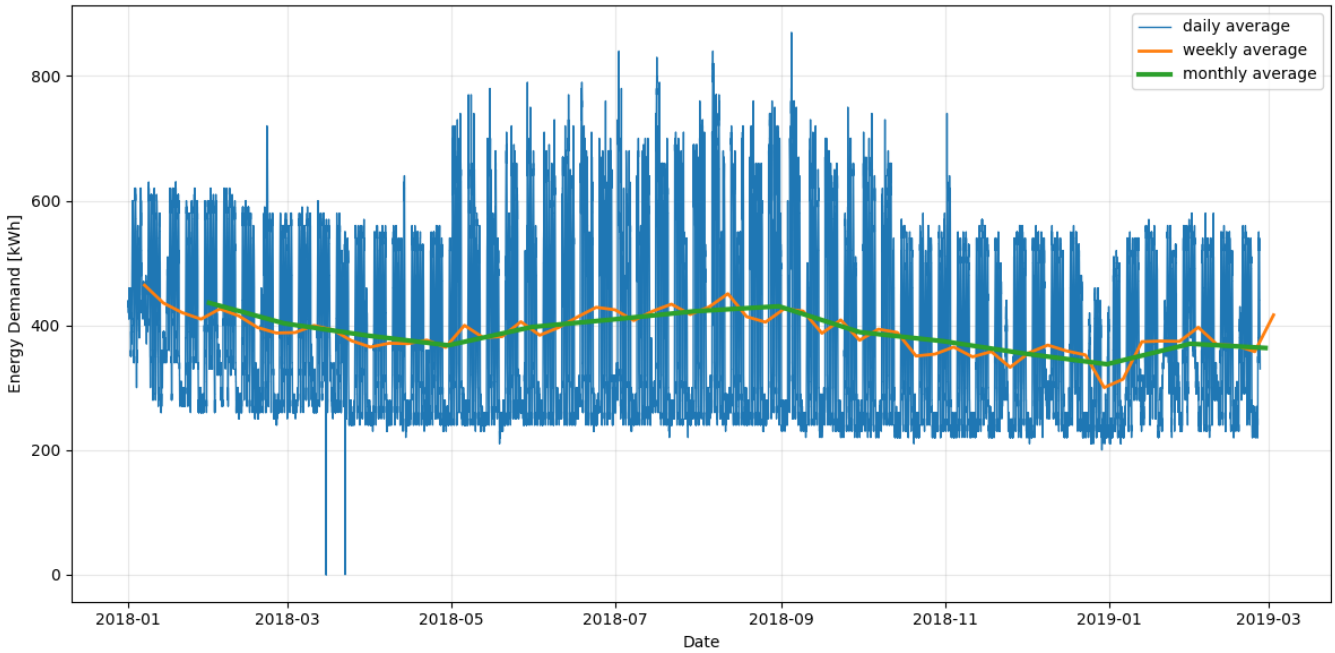


- Spring (March–May) : Lowest demand period overall

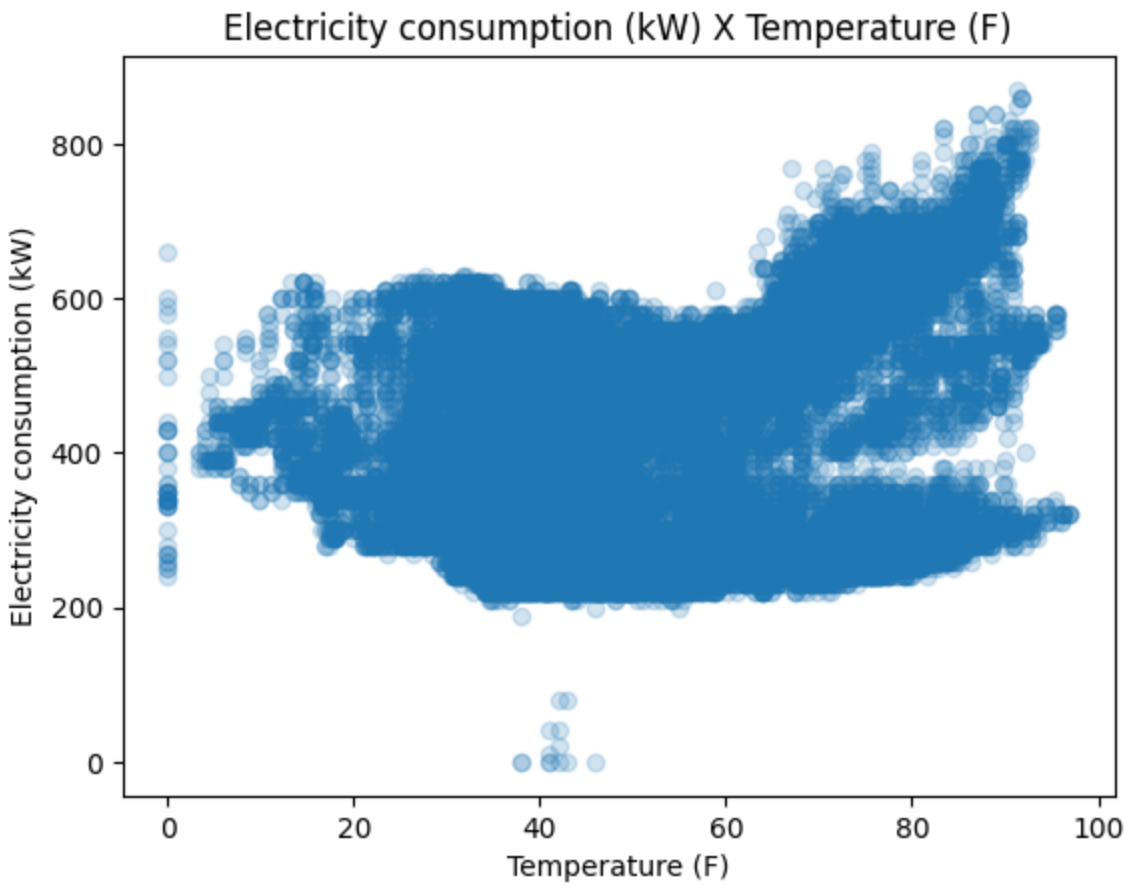
- Summer (June–August) : Peak demand season
- Autumn (September–November) : Declining demand transitioning from summer peaks
- Winter (December–February) : Moderately elevated demand compared to spring

Now, let's consider all of this information in a singe graph.

The following graph, give us relevant information about trending. Daily demand is highly volatile throughout, with regularly spanning 200–870 kWh, though weekly averages smooth this to a consistent seasonal signal. A notable anomaly appears in March 2018, where demand briefly dropped near zero, likely indicating a data gap or outage event.

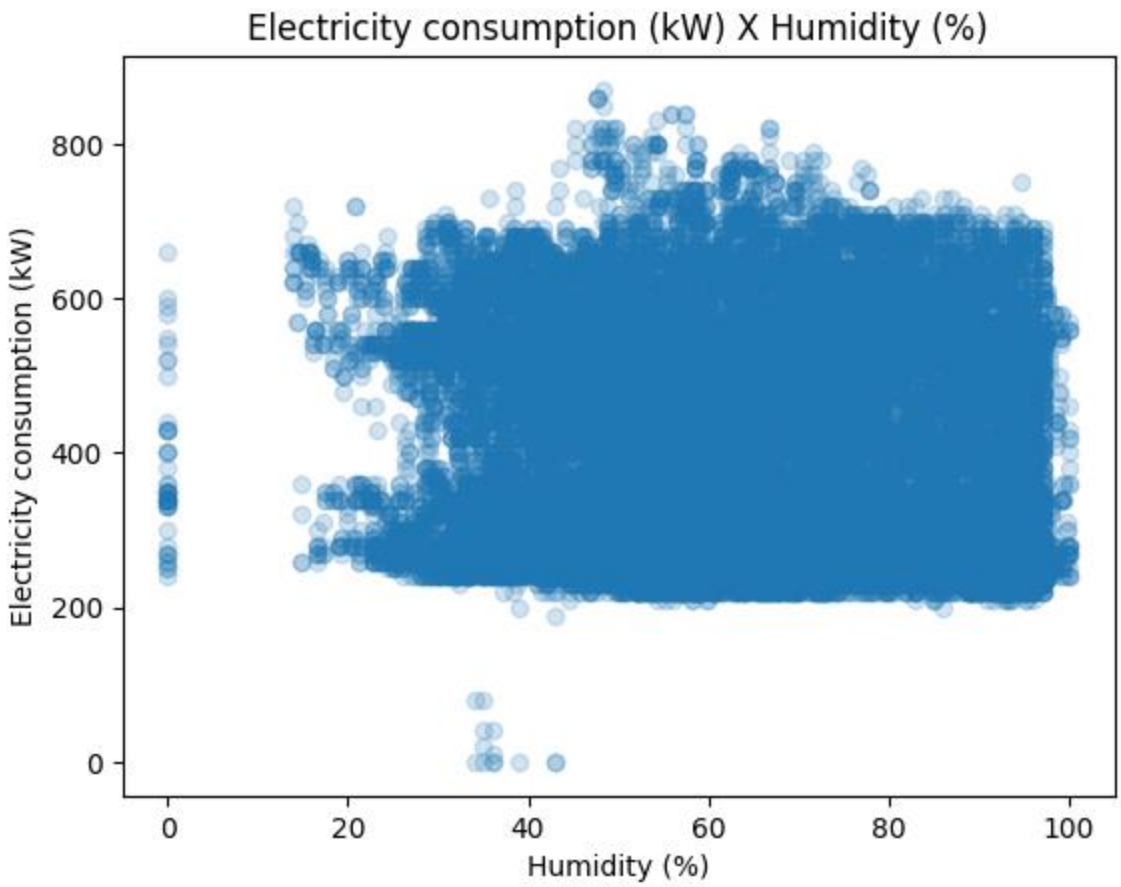


1.c) What relationship do you observe between electricity demand and weather variables ?



The scatter plot of electricity demand versus outdoor temperature suggests the existence of a comfort temperature (typically ~55–65°F) around which HVAC-related load is minimal.
Below this reference temperature, electricity consumption increases due to heating demand, while above it, cooling demand dominates.

This behavior indicates a non-linear relationship between electricity demand and temperature, which can be effectively modeled using a non-linear model. Furthermore, on the top of plot, we see a form such as draw a U or V-shaped. That precious information will be use in features engineering parts.

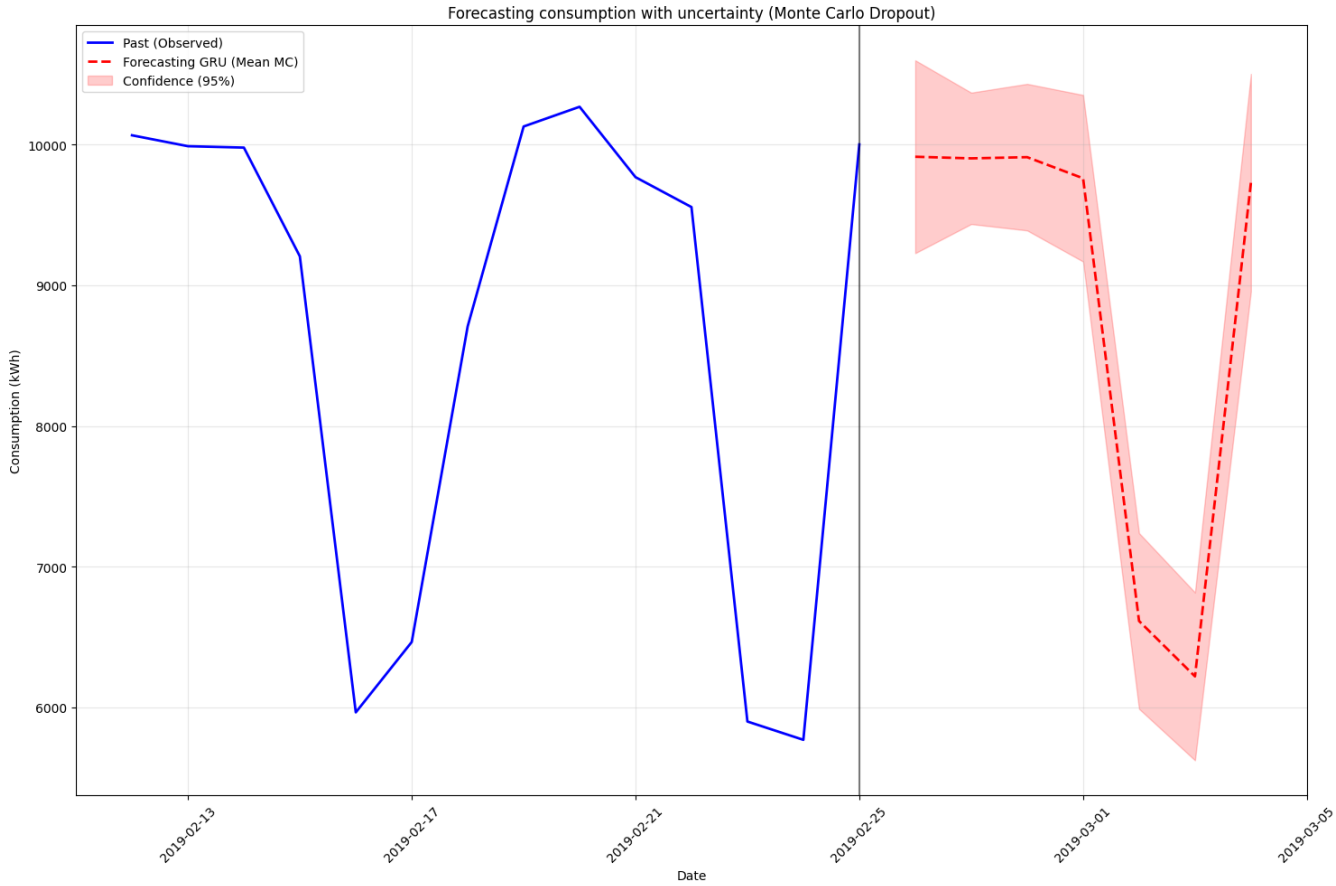


Large vertical dispersion for almost all humidity values. For the same humidity value, consumption varies significantly at constant humidity

We assumed Humidity alone cannot explain the variations in consumption.

2. Forecasting electricity consumption:

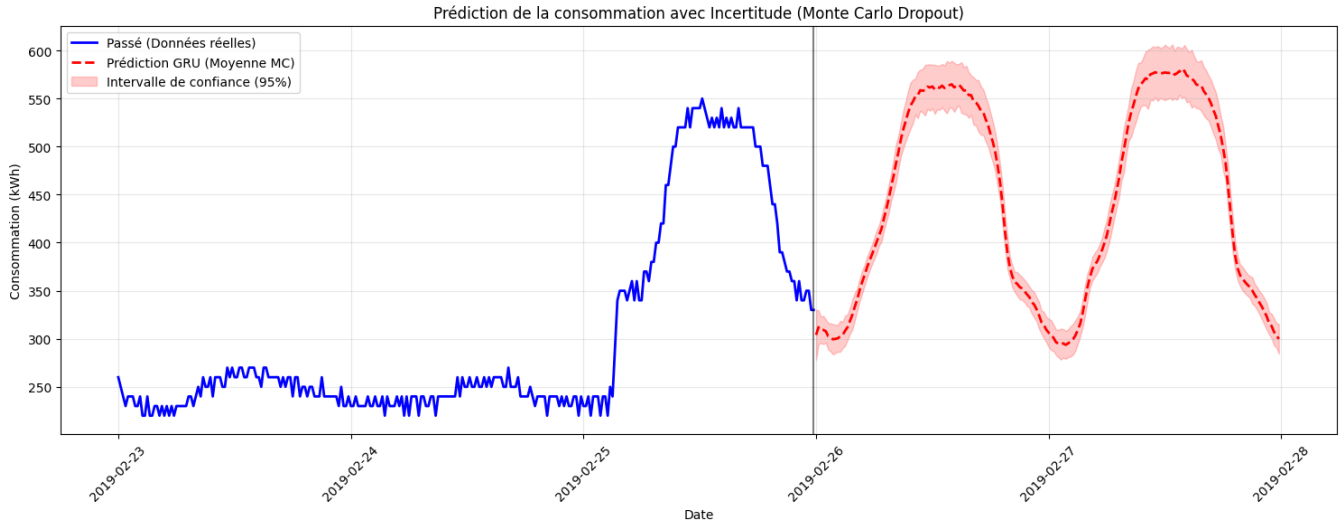
a.i) Forecast the electricity consumption (KwH) for the next 7 days



Cyclical Consistency: The model seems to have learned the periodicity of the data. It predicts a significant drop followed by a sharp recovery, which aligns with the behavior seen around February 17th and February 24th.

a.ii) Forecast the electricity consumption (Kw) for the next 2 days

This is high-resolution, granular data. The model must capture intra-day patterns (morning spikes, lulls in the afternoon, evening peaks). It deals with "volatility" (sudden changes) much more than the daily forecast does.



The Forecast (Red Dashed): The model predicts that this high-usage pattern will continue. It forecasts two distinct daily peaks reaching approximately 575 kWh on February 26th and 27th.

Confidence Interval (Pink Shaded): The 95% confidence interval is tightest during the "slopes" (as consumption rises or falls) and widest at the peaks and troughs. This indicates the model is more certain about the direction of change than the exact maximum or minimum values.

b) Feature Selection

The features selected for each model reflect the known physical, behavioural, and temporal drivers of energy use in a commercial building. They are organised into four logical categories, each targeting a distinct dimension of the forecasting problem.

Electricity Demand Model

The following features were engineered and selected to capture the dominant drivers of instantaneous power draw (kW) in the building:

Category	Feature(s)	Purpose & Logic
1. Autoregressive (Memory)	Electricity consumption (kW)	Provides the base target variable to establish the current scale of usage.
	lag_1, lag_2, lag_96	Captures immediate momentum (1-2 steps) and daily seasonality (96 steps / 24h).
	energy_kW_rolling_4	Smooths high-frequency noise to help the GRU identify broader trends over spikes.
2. Weather & Thermodynamics	Temperature (F), Humidity (%)	Acts as the primary external driver for HVAC (Heating, Ventilation, Air Conditioning) loads.
	temp_rolling_12, humid_rolling_12	Accounts for thermal mass, capturing accumulated heat and moisture over a 3-hour window.
	HDD_occupied, CDD, Temp_Squared	Represents energy needed for climate control and accounts for non-linear weather-to-energy relationships.
3. Temporal & Cyclical	hour_cos, month_cos	Prevents mathematical discontinuities between hour 23 and 0 by mapping time to a circular continuum.
	is_summer, is_winter	Seasonal switches that help the model shift between heating-heavy and cooling-heavy logic.
	is_weekend, is_friday, is_monday	Flags behavioral changes on weekends and bridge days (Fridays/Mondays).
4. Operational Context	is_occupied	Captures the drastic shift in energy profile caused by lighting, computers, and active ventilation.

Electricity Consumption Model

For interval-level energy consumption (kWh), a complementary but distinct feature set was constructed, placing greater emphasis on thermal lag and degree-day accumulation:

Category	Feature(s)	Purpose & Logic
1. Memory & Smoothing	energy_interval_kWh	Serves as the primary target and baseline for the next-step prediction.

Category	Feature(s)	Purpose & Logic
	energy_interval_kWh_rolling_3	Averages the last three intervals to filter noise and focus the GRU on the general load level.
	lag_1, lag_7	lag_1 captures immediate momentum; lag_7 identifies patterns from the same time in previous cycles.
2. Weather Dynamics	Temperature (F), Humidity (%)	Accounts for HVAC load; specifically, how rising humidity forces systems to dehumidify and cool.
	temp_rolling_3, humid_rolling_3	Accounts for thermal lag, representing the time it takes for outdoor heat to penetrate building walls.
	Temp_Squared	Models the non-linear U-shaped energy curve where consumption spikes at both temperature extremes.
	DJC & DJF	Represents Cooling and Heating Degree Days (thermal debt) needed to maintain a comfortable indoor setpoint.
3. Calendar & Cyclical	is_weekend, is_friday, is_monday	Binary flags that signal the model to shift logic for different operational schedules (bridge days).
	is_winter	A seasonal switch that allows the model to prioritize heating-related features over cooling ones.
	month_cos/sin, day_cos/sin	Trigonometric encoding treats time as a continuous circle rather than a linear jump.

c) Model Performance

Model performance is assessed through two complementary methods: (1) statistical metrics computed on training and test sets, and (2) a visual analysis of prediction error distributions. Together, these approaches provide both a quantitative summary and a qualitative understanding of model behaviour.

Evaluation Metrics

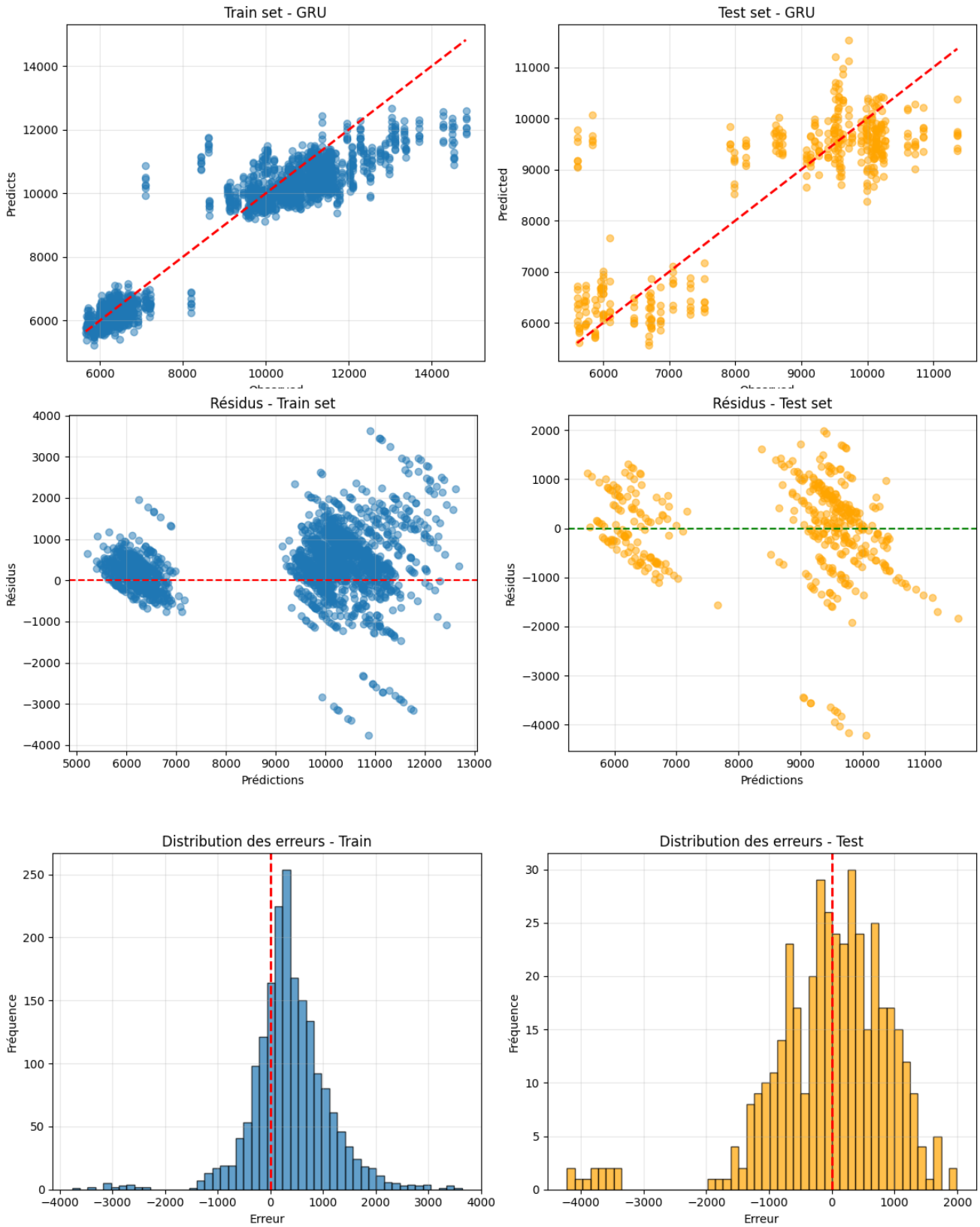
Four standard regression metrics are used to quantify the distance between predicted values ($\hat{y}$) and observed values (y):

Metric	Definition & Role in Energy Forecasting
MAE	Mean Absolute Error — average absolute difference between predicted and actual values. Provides a linear, equally-weighted score.
MSE	Mean Squared Error — squares each error term, penalising large outliers more heavily. Critical in energy forecasting where massive under-predictions could lead to grid instability.
RMSE	Root Mean Squared Error — square root of MSE, bringing the metric back to original units (kWh). Easier to interpret relative to the building's average load.
R^2	Coefficient of Determination — measures how well the model explains data variance. R^2 close to 1.0 indicates the GRU successfully captured seasonality, weather, and occupancy patterns.

Results — Electricity Consumption Model

Metric	Train Set	Test Set	Delta	Interpretation
MSE	697,984.83	959,201.63	+261,216.80	Large increase indicates significant misses on unseen data, likely during peak consumption spikes.
RMSE (kWh)	835.45	979.39	+143.94	On new data, the model deviates by ~980 kWh. The 17% increase from train to test suggests moderate overfitting.
MAE (kWh)	607.70	705.82	+98.12	Since MAE < RMSE, the model is generally accurate, but a few large errors skew the results.
R^2 Score	0.8625	0.6760	-0.1865	The model captures 86% of patterns in training but only 68% on test data, leaving 32% of real-world variance unexplained.

Visualizion – Analysis of errors



Analysis of prediction error distributions reveals three important phenomena:

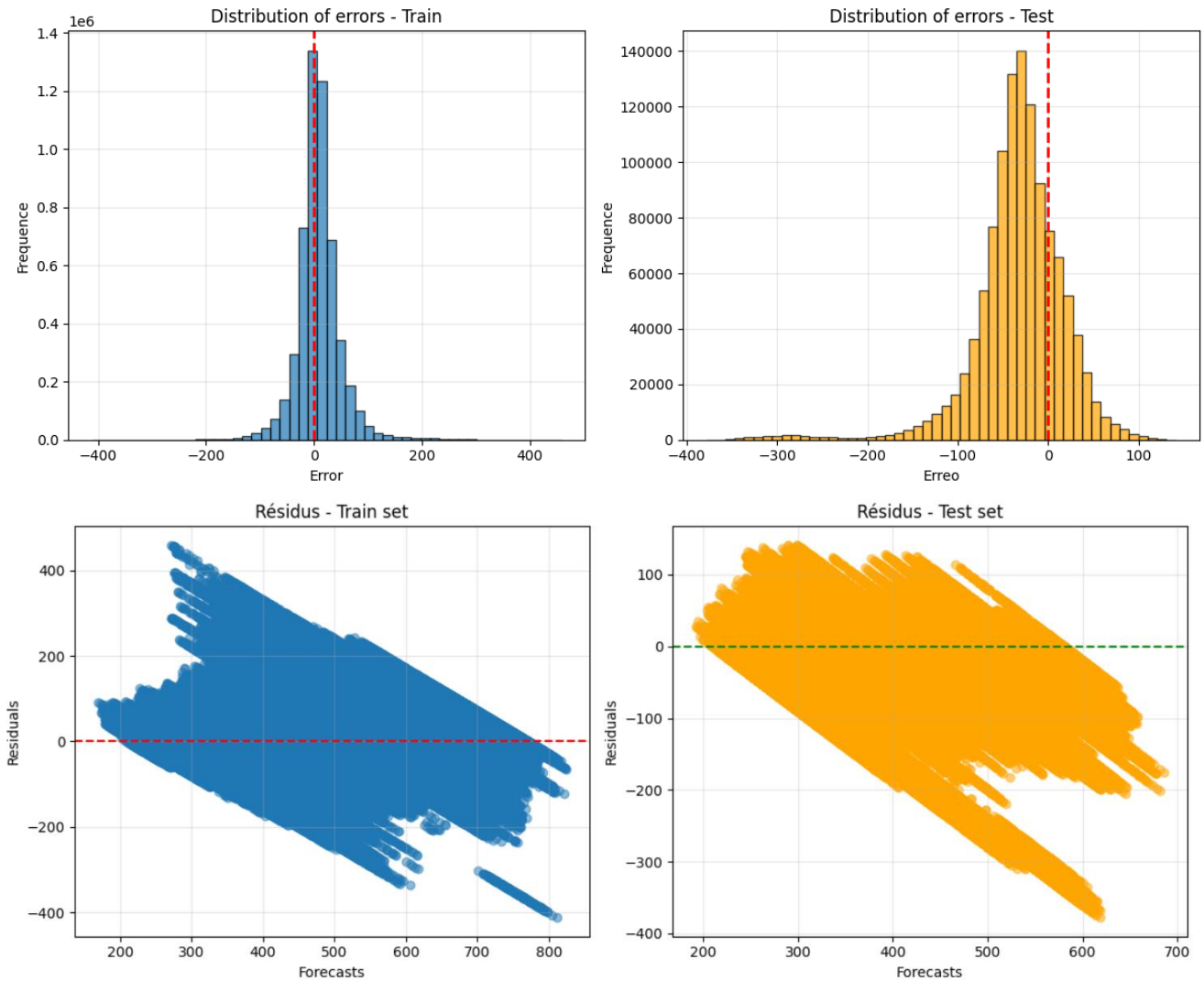
1. **Centred residuals on training data:** The near-zero mean of the training error distribution confirms that the model is correctly specified and captures the central tendency of electricity consumption without systematic bias.
2. **Wider error spread on test data (R^2: 0.86 → 0.67):** The broadening of the test error distribution is consistent with three phenomena: partial memorisation of training noise (overfitting), mild non-stationarity between the two periods, and heteroscedasticity, where forecast error increases during consumption peaks.
3. **Systematic outliers (−3,000 to −4,000 kWh):** A right-side fat tail in the error histogram indicates that the model systematically underestimates high-energy spikes, a known limitation of smooth recurrent architectures in the presence of sudden load changes.

Results — Electricity Demand Model

Metric	Train Set	Test Set	Interpretation
MSE	1,698.38	3,651.49	Penalizes large errors; the jump to test indicates more frequent misses on new data.
RMSE	41.21	60.43	On average, predictions deviate by approximately 60 kWh in the test phase.
MAE	27.79	43.92	The absolute average error is roughly 44 kWh.
R^2 Score	0.925	0.701	The model explains 92.5% of variance during training, but only 70% on test data, revealing generalisation limits.

The demand model achieves a strong R^2 of 0.925 on training data, demonstrating that the GRU successfully captures the dominant patterns in instantaneous power demand. The degradation to 0.701 on the test set is consistent with the consumption model and reflects the same combination of moderate overfitting and temporal non-stationarity. The RMSE of 60.43 kW on unseen data represents an average deviation of approximately 60 kW from actual demand a performance level that is adequate for operational energy management while highlighting room for improvement during extreme load conditions

Visualizion – Analysis of errors



Both the Train (blue) and Test (orange) plots, we see distinct downward-sloping diagonal lines. This usually happens in energy models when the model is "missing" a categorical factor (like a specific shift change or a binary switch).

Constant Error: These lines suggest that for certain predicted values, the actual value is consistently off by a fixed amount.

Heteroscedasticity: The spread of errors changes as the forecast value increases. The model is more precise at low consumption levels (~200 kWh) but becomes more "confused" and scattered at high consumption levels (600+ kWh).

d) Add features

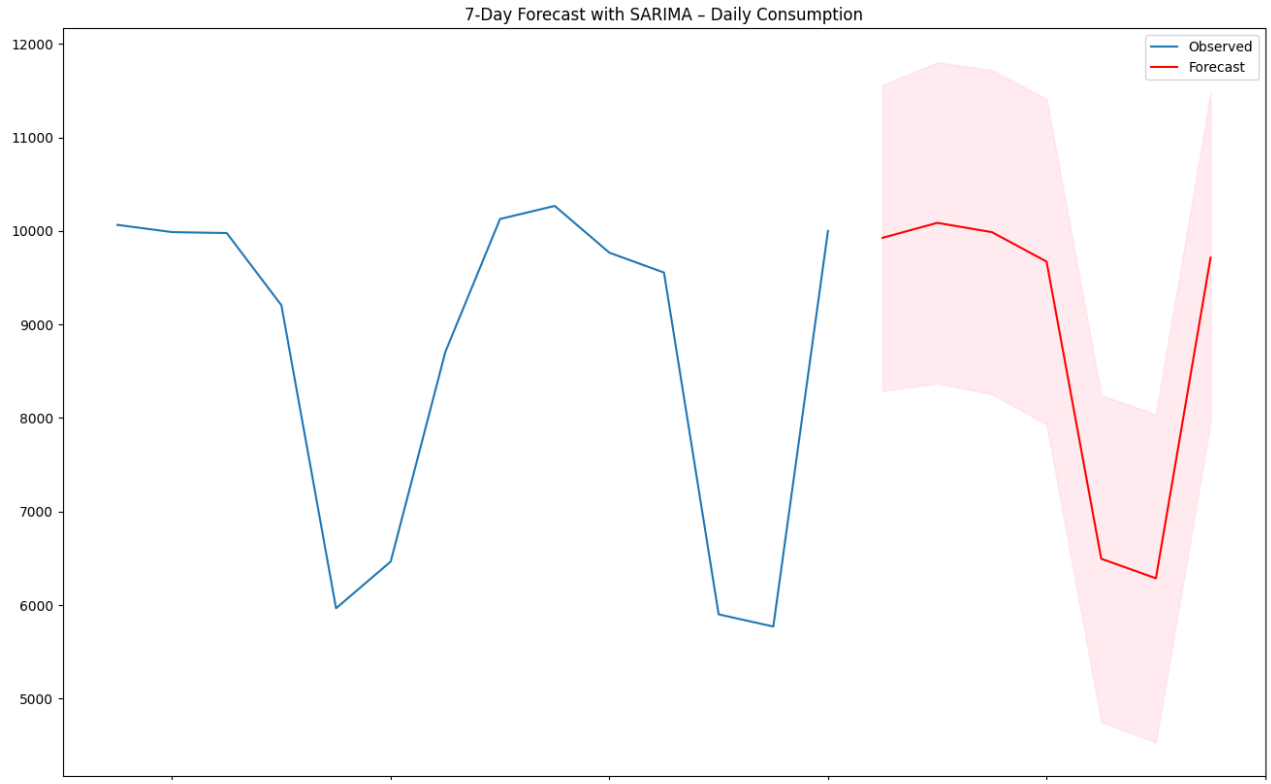
Solar Radiation: This provides a measure of "Solar Gain." Even if the temperature is the same, a sunny day will heat a building significantly more than a cloudy one due to radiation through windows. This could help the model better predict the HVAC cooling spikes you are currently under-predicting.

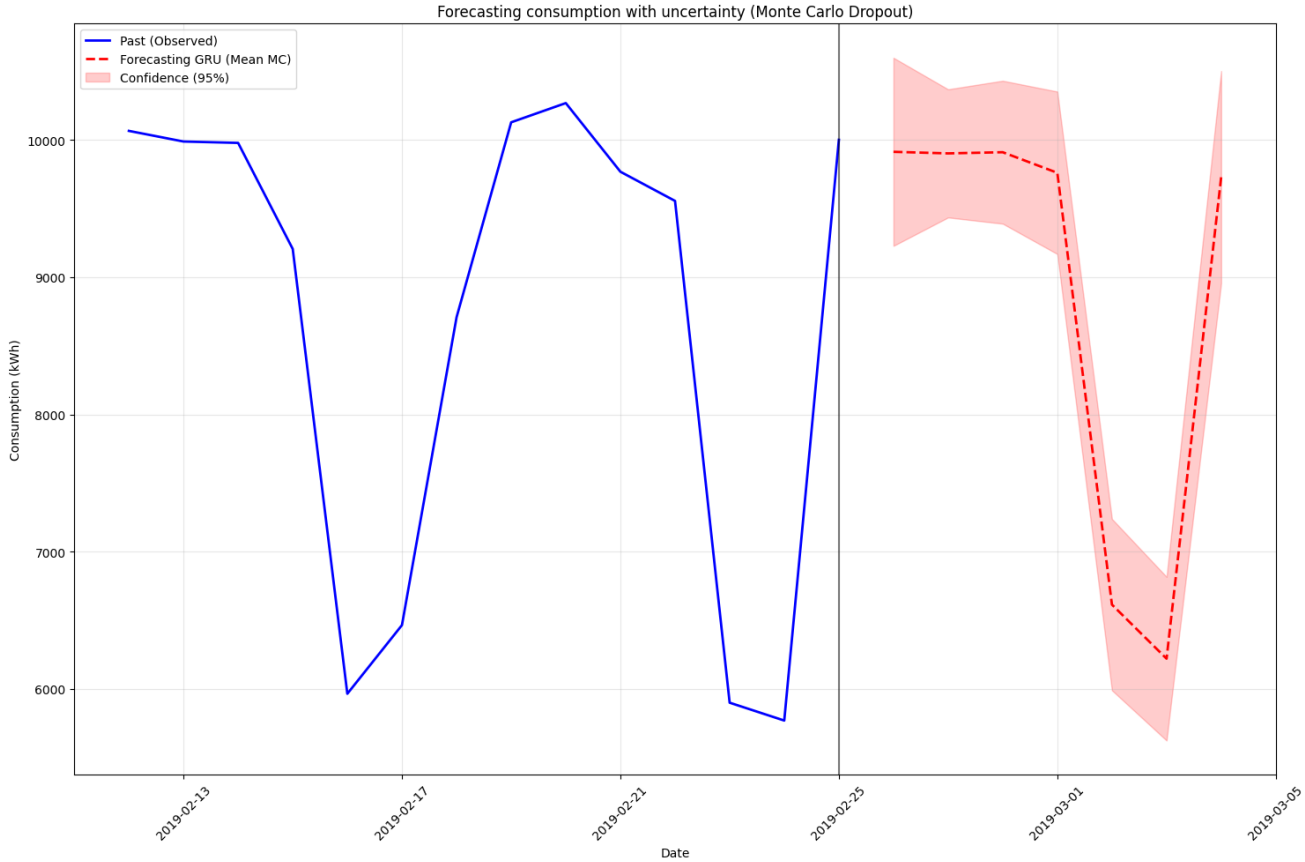
Weather Forecasts (vs. Historicals) : Currently, your model uses historical weather (temp_rolling). In a real-world deployment, the model needs the forecasted temperature for the next 24 hours to plan its energy

prediction. Using "Predicted vs. Actual" weather features helps the model account for the uncertainty in the weather report itself.

4. Bencharmaking : SARIMA vs Seq2Seq

Let’s have a quick how Seq2Seq outperforming SARIMA to handle good forecasting and better confidence band.


7-Day Forecast with SARIMA – Daily Consumption
Observed
Forecast
12000
11000
10000
9000
8000
7000
6000
5000


Forecasting consumption with uncertainty (Monte Carlo Dropout)
Past (Observed)
Forecasting GRU (Mean MC)
Confidence (95%)
Consumption (kWh)
10000
9000
8000
7000
6000
2019-02-13
2019-02-17
2019-02-21
2019-02-25
2019-03-01
2019-03-05
Date